INCH-POUND
MIL-DTL-16377/61A(SH)
2 November 2021
SUPERSEDING
MIL-F-16377/61(SH)
14 May 1982

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT AND SOLID-STATE EQUIVALENT, FLOODLIGHTS, 300 WATTS, 120 VOLTS, DRIP-PROOF 15 DEGREES, SYMBOLS 303.2, 303.3, 303.4,303.2SSL, 303.3SSL, AND 303.4SSL

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 7 provide information for the light fixtures covered by this specification sheet. See the Requirements sections for specific references to these figures.

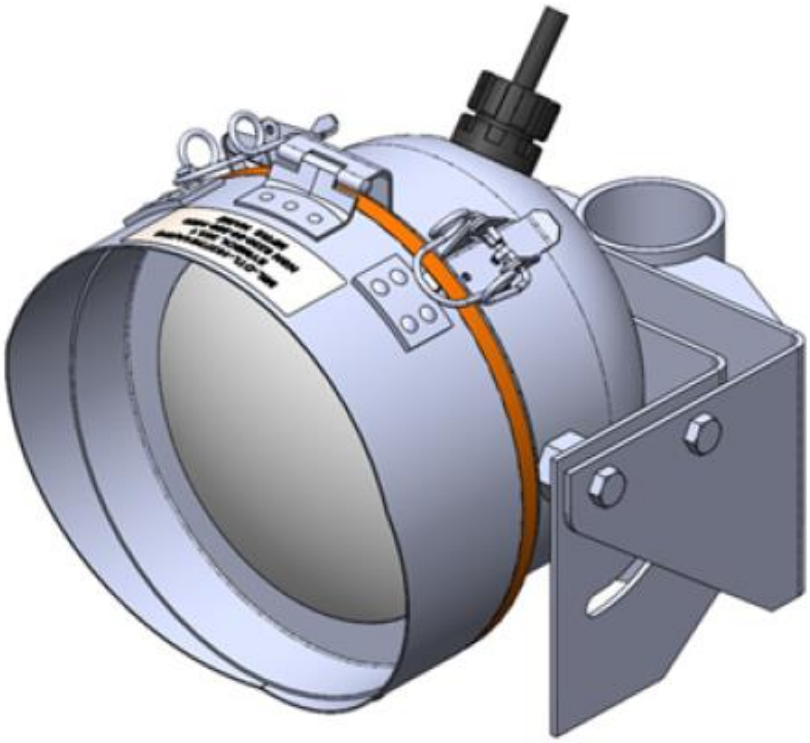


FIGURE 1. Typical view of symbols 303.2, 303.3, and 303.4.

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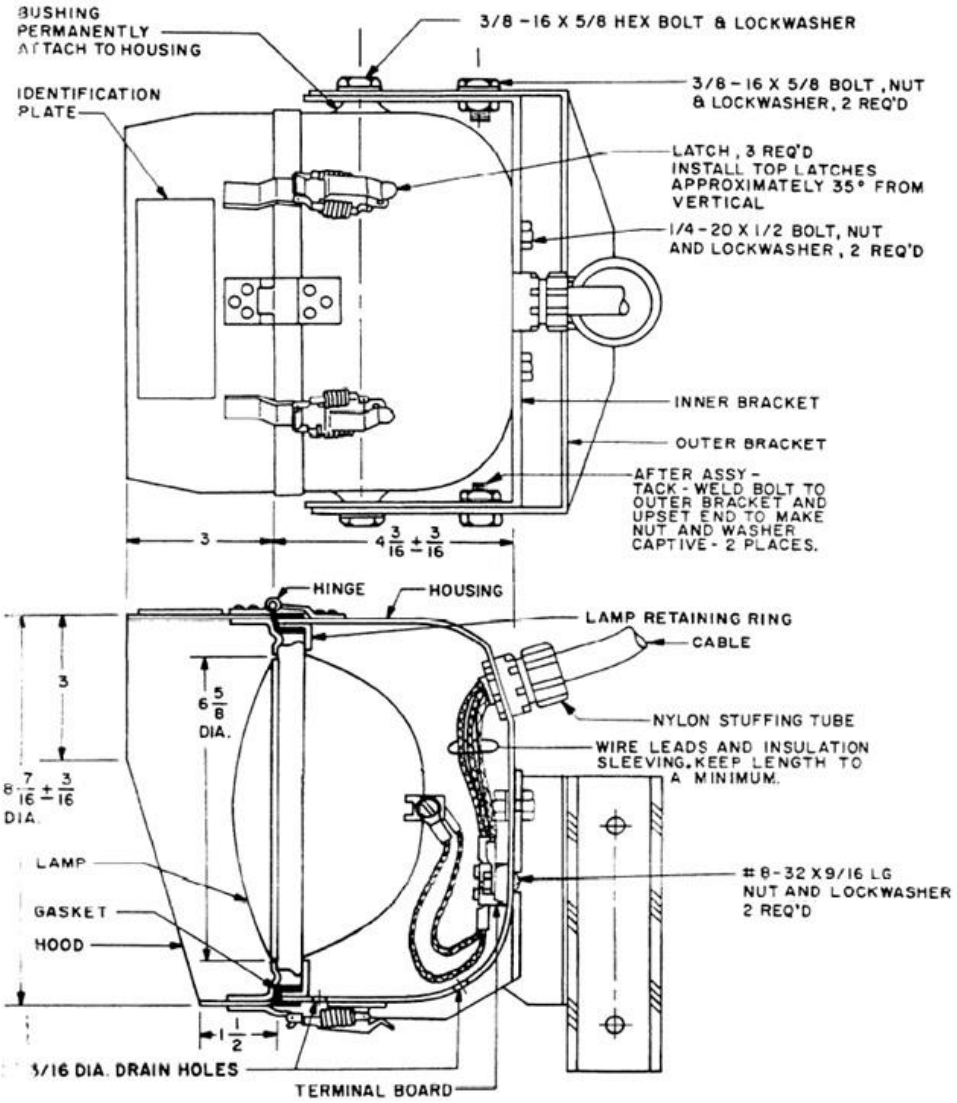


FIGURE 2. Typical top and section view, incandescent.

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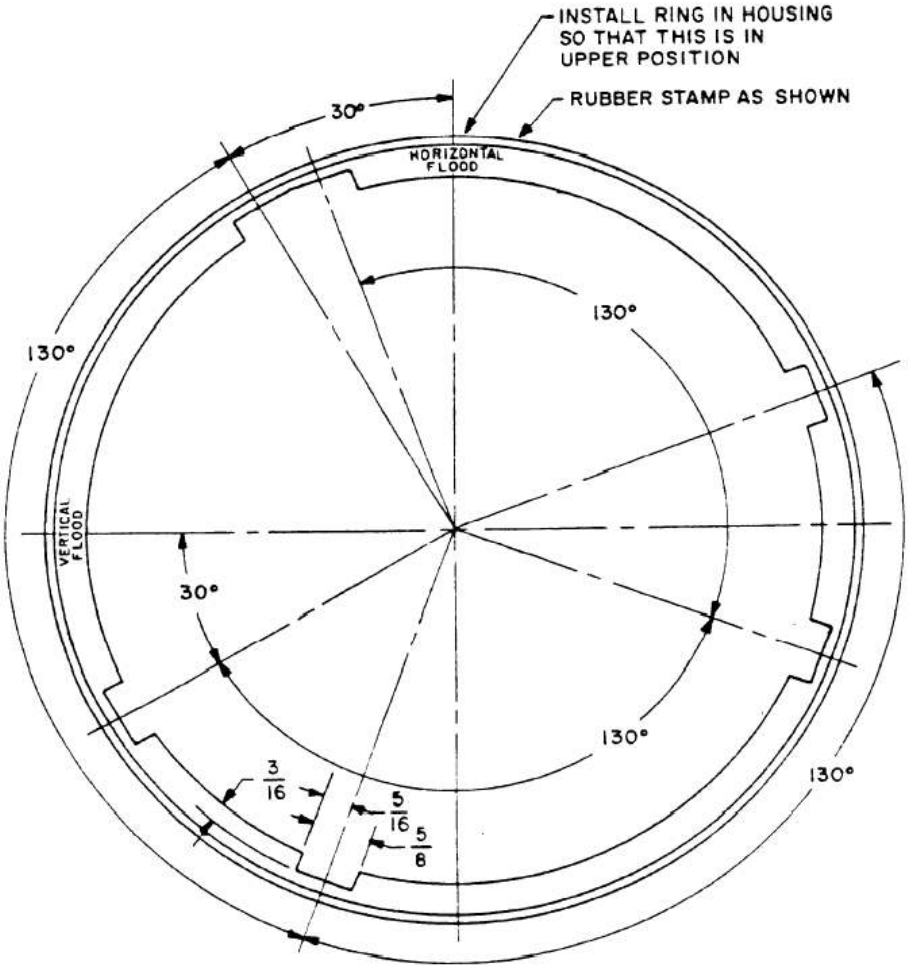


FIGURE 3. Lamp retaining ring, incandescent.

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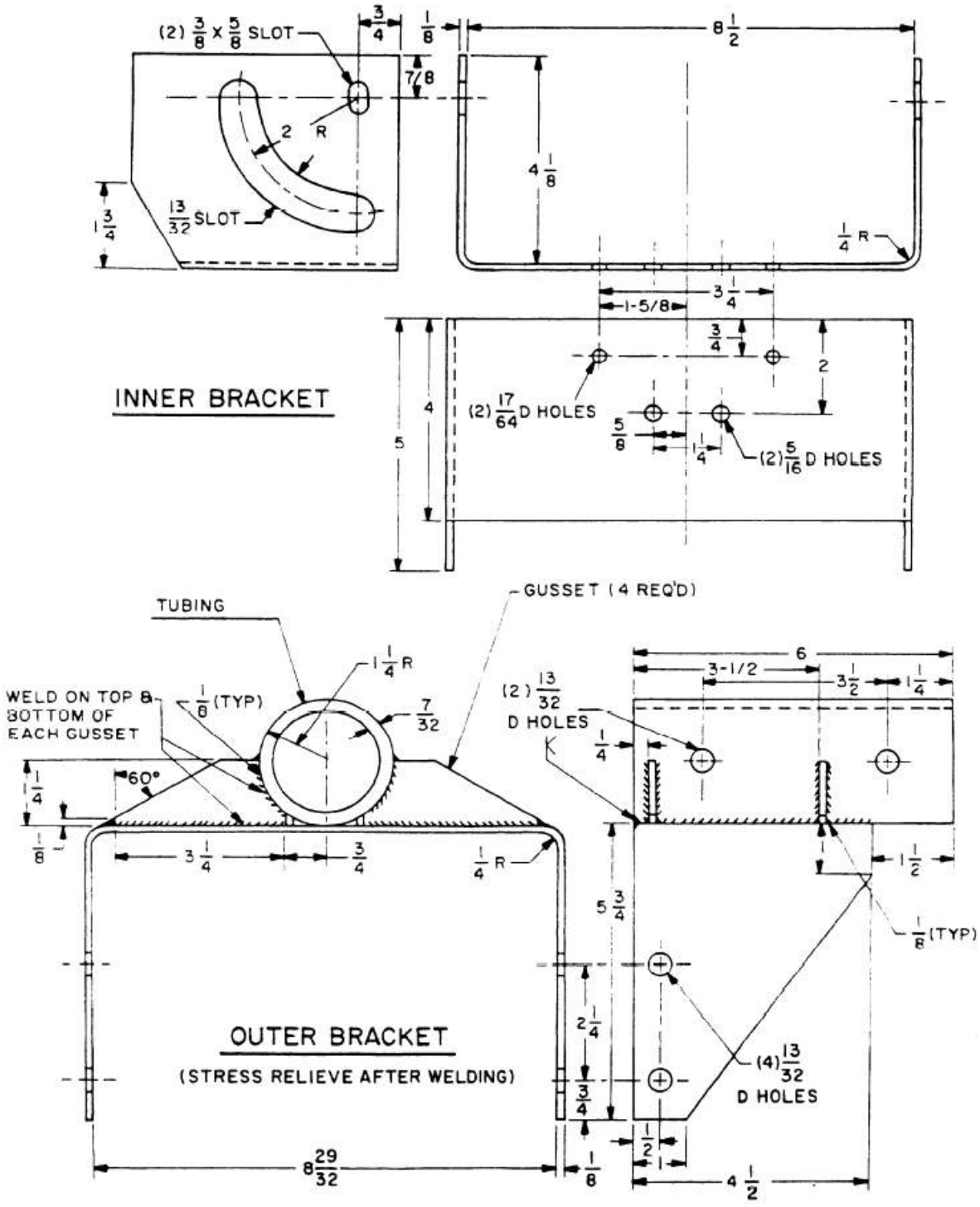


FIGURE 4. Inner and outer brackets, incandescent.

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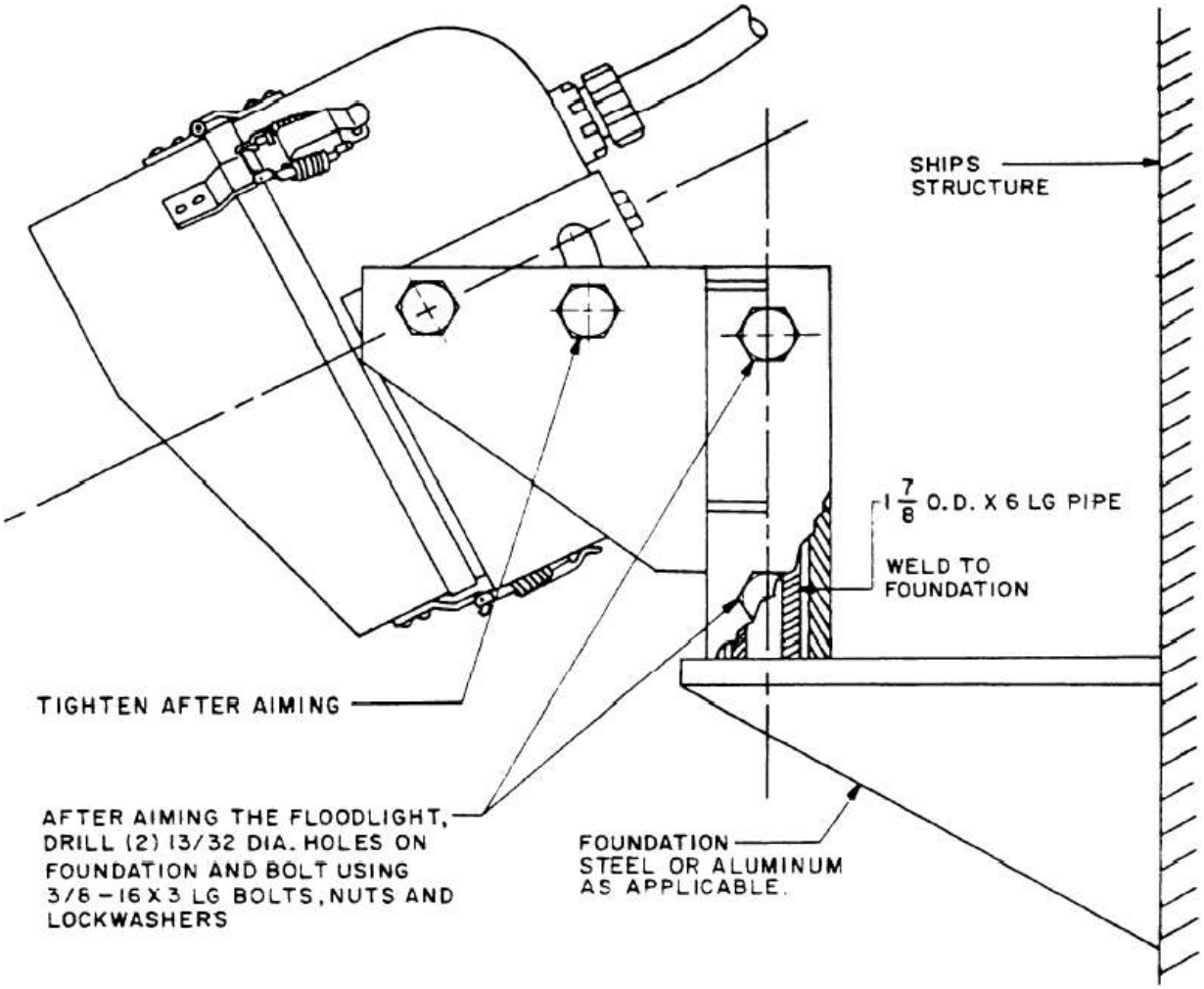


FIGURE 5. Typical mounting detail.

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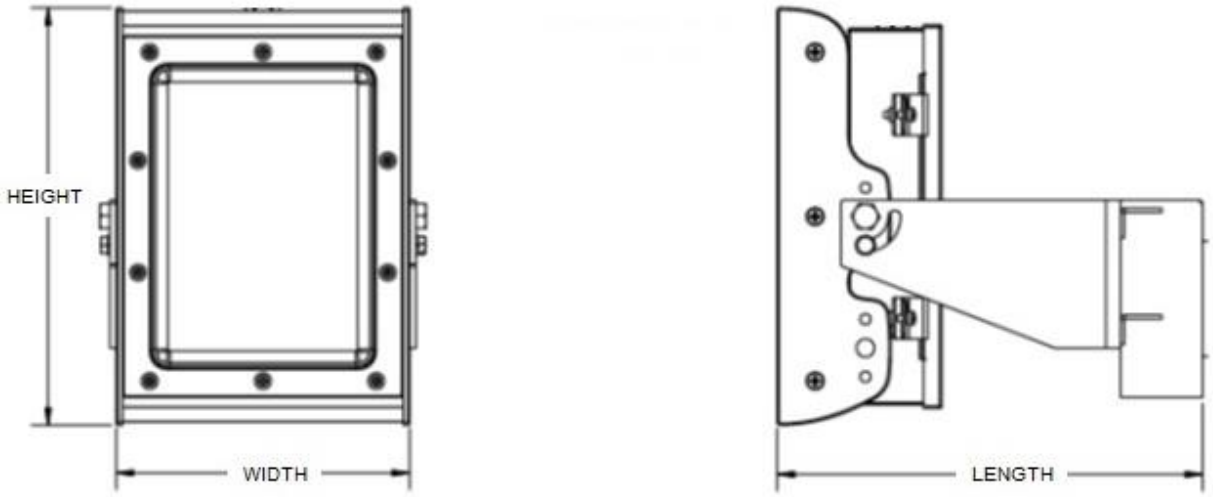
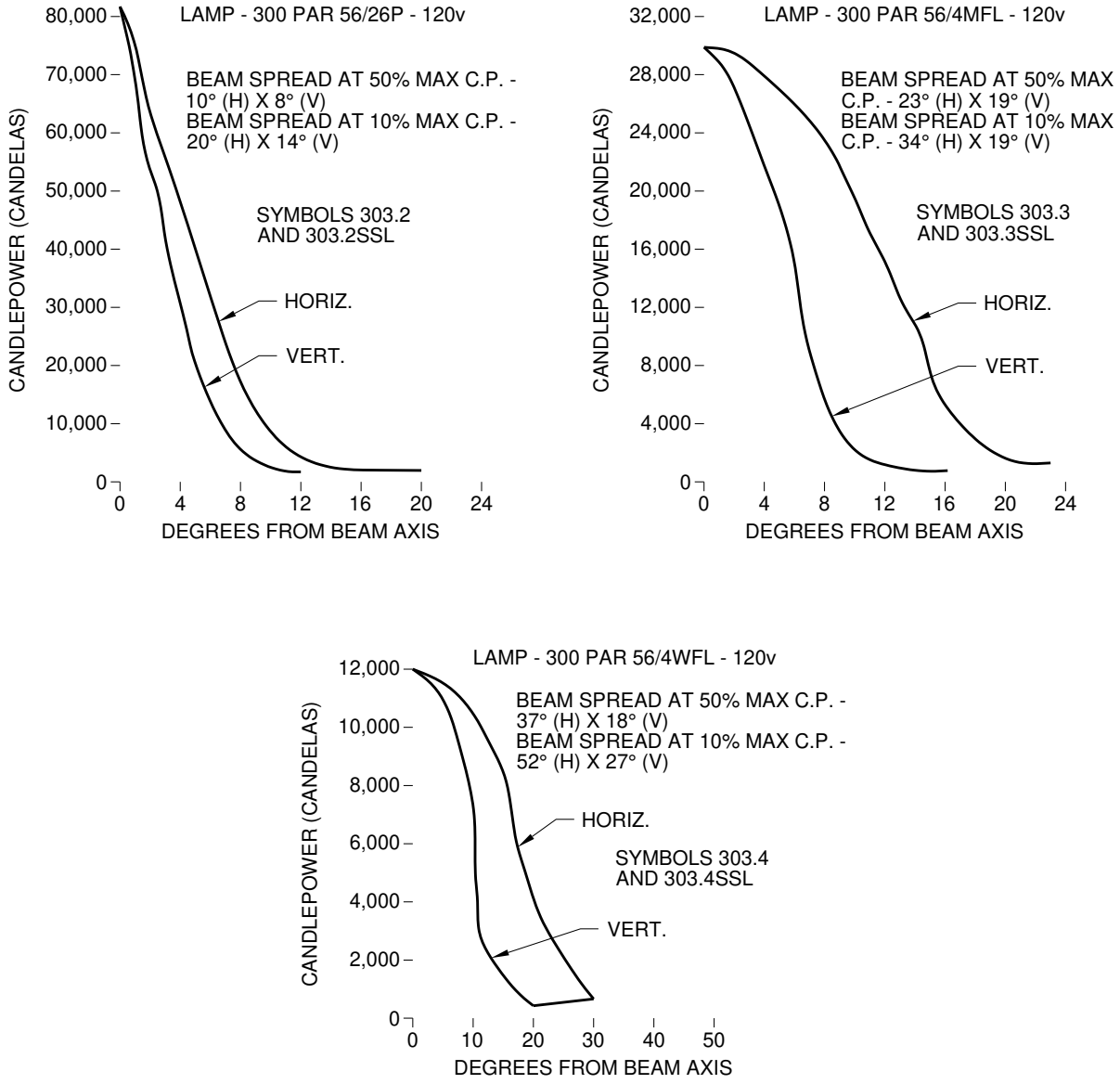


FIGURE 6. Typical solid-state lighting (SSL) fixture.

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NOTE: Lamp designations are for incandescent fixtures only.

FIGURE 7. Typical candlepower distribution.

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INCANDESCENT REQUIREMENTS

CLASSIFICATION

Type: II
Class: 2

SYMBOL NUMBERS

303.2: Floodlight fixture for PAR56 300-watt, 120-volt spot lamp, Industry Number 300PAR56/2SP, white.
303.3: Floodlight fixture for 300-watt, 120-volt medium flood lamp, Industry Number 300PAR56/4MFL, white.
303.4: Floodlight fixture for 300-watt, 120-volt wide flood lamp, Industry Number 300PAR56/4WFL, white.

DESIGN

Electrical rating:	300 watts, 120 volts
Dimensions and configuration:	See figures 1 through 5. Housing dimensions may be considered to be variable to the extent that available commercial tooling can be used.
Material:	Corrosion-resistant steel (CRES) 316, except as otherwise indicated in the specification. Material for lamp retaining ring may be phosphor bronze.
Enclosure:	Drip-proof, 15 degrees
Lamp:	See general information in table III. Lamp shall not be furnished.
Cable:	Floodlights shall be furnished with 10 feet of 3-conductor low smoke cable, type LSTHOF-3, in accordance with MIL-DTL-24643/3. The third conductor shall be used for grounding. The length of the wire leads to the terminal board and ground shall be kept to a minimum. Terminal lugs (SAE AS17143) shall be used.
Insulation sleeving:	The entire length of the cable wire leads to the terminal board and internal wiring shall be fitted with insulation sleeving conforming to MIL-I-3190/6.
Internal wiring:	The enclosure shall be furnished with two wire leads, 15 inches long each, attached to the terminal board. These leads are for connection to the lamp, and both ends shall be fitted with terminal lugs conforming to SAE AS17143. The wire leads shall be #16 AWG, stranded, 250 °C application, nickel-clad copper, Teflon, or equal, jacket with overbraid of silicone glass braid dip.
Terminal board:	The terminal board shall be type 8TB-02F in accordance with A-A-59125.
Nylon stuffing tube:	The nylon stuffing tube shall be in accordance with ASTM F1836M. The packing assembly and “O” ring shall be as specified in ASTM F1836M for size 2 tube.
Weight:	14 pounds, maximum
Designation and marking:	A type I identification (ID) plate shall be furnished in accordance with MIL-DTL-16377. Location of the ID plate shall be as shown on figure 2. ID plates shall be of metal material and corrosion-resistant in accordance with MIL-DTL-16377.

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SSL REQUIREMENTS

CLASSIFICATION

Type: III
Class: 2

SYMBOL NUMBERS

DESIGN

303.2SSL: Floodlight fixture, spot, Solid State Source, white.
303.3SSL: Floodlight fixture, medium flood, Solid State Source, white.
303.4SSL: Floodlight fixture, white flood, Solid State Source, white.

Electrical rating:	150 watts, 120 volts
Maximum dimensions:	Length, 15 inches (including mounting hardware); width, 15 inches; height, 10 inches. Mounting shall be such that fixtures can be rotated continuously from full horizontal to full vertical (see figure 6).
Photometric data:	See figure 7 of this sheet and 3.7.1 of MIL-DTL-16377. The minimum luminescence shall be not less than that shown on figure 7.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.
Enclosure:	Drip-proof, 15 degrees
Cable:	Floodlights shall be furnished with 10 feet of 3-conductor low smoke cable, type LSTHOF-3, in accordance with MIL-DTL-24643/3. The third conductor shall be used for grounding.
Nylon stuffing tube:	The nylon stuffing tube shall be in accordance with ASTM F1836M.
Weight:	20 pounds, maximum
Designation and marking:	A type I ID plate shall be furnished in accordance with MIL-DTL-16377. ID plates shall be of metal material.
Material:	Corrosion-resistant in accordance with MIL-DTL-16377.

EXAMINATION AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete floodlights	See tables I and II.
Conformance:	
Comparison inspection:	Same as the first article inspection.
Inspection of product for delivery:	
Examination:	See MIL-DTL-16377.
Examination for preparation of delivery:	See MIL-DTL-16377.
Quality conformance tests:	See MIL-DTL-16377.

GENERAL INFORMATION

Assignment of Military part numbers and general information is shown in table III.

For incandescent, this floodlight is designed so that a typical lamp, having a beam spread of 60° horizontal, 30° vertical, can be repositioned in the floodlight to obtain a reverse beam spread of 30° horizontal, 60° vertical by rotating the lamp 90° in either direction.

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TABLE I. First article and comparison inspection for complete incandescent floodlights, symbols 303.2, 303.3, and 303.4.

Test	Requirement paragraph of MIL-DTL-16377	Test paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	--
Operation	3.13	4.8.1	--
Dielectric withstanding voltage	3.6.15	4.8.2	--
Insulation resistance	3.6.16	4.8.3	--
Enclosure effectiveness	3.5.11	4.8.14	Drip-proof, 15 degrees 1/
Shock	3.5.12	4.8.7	Grade B 1/
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14	Drip-proof,15 degrees 1/
Dielectric withstanding voltage	3.6.15	4.8.2	--
Insulation resistance	3.6.16	4.8.3	--
Grounding circuit	3.6.13.3	4.8.11	--
Salt fog	3.5.15	4.8.10	--

NOTE:

1/ Floodlight for this test shall be mounted in a manner similar to that shown on figure 5.

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TABLE II. First article and comparison inspection for complete type III fixtures, symbols 303.2SSL, 303.3SSL, and 303.4SSL.

Test	Requirement paragraph of MIL-DTL-16377	Test paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Enclosure effectiveness	3.5.11	4.8.14	Drip-proof, 15 degrees 1/, 2/
Shock	3.5.12	4.8.7	Grade B 1/, 2/
Vibration	3.5.13	4.8.8	1/, 2/
Enclosure effectiveness	3.5.11	4.8.14	Drip-proof, 15 degrees 1/, 2/
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Photometrics	3.7.1	4.8.6	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Failsafe circuit design	3.17	4.8.24	
Stress relief	3.8.5	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Salt fog	3.5.15	4.8.10	

NOTES:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

2/ The floodlight for this test shall be mounted in a manner similar to that shown on figure 5.

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TABLE III. Assignment of part numbers and general information.

Symbol number and Military part number M16377/61-	Item name
303.2	Symbol 303.1 with 300 watts, 120 volts spot lamp, Industry Number 300Par56/2SP, MS15535-4; Beam spread: 20° horizontal, 15° vertical.
303.3	Symbol 303.1 with 300 watts, 120 volts medium flood lamp, Industry Number 300PAR56/4MFL; Beam spread: 35° horizontal, 20° vertical.
303.4	Symbol 303.1 with 300 watts, 120 volts wide flood lamp, Industry Number 300PAR56/4 WFL, MS15535-6; Beam spread: 60° horizontal, 30° vertical.
303.2SSL	SSL spot lamp with minimum spot lamp candlepower as shown on figure 6.
303.3SSL	SSL medium lamp with minimum medium lamp candlepower as shown on figure 6.
303.4SSL	SSL wide lamp with minimum wide lamp candlepower as shown on figure 6.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6230-2021-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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